

Indoor Navigation for the Visually Impaired (VI) using QR Code

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ABSTRACT

This project is regarding a development-based project on an indoor navigation system for visually impaired (VI) users. The system aims to help the VI users move from one location to another within the indoor building by scanning the QR codes on the door glass using a phone camera scanner. The test scene would be our FICT department, which is Block N. In order to deploy the system incrementally, the system will consist of four main modules:

- a) The QR code detection module detects the color of QR code and the angle
- b) The QR code reader module decodes the user's current location on a map
- c) The route guidance module computes an optimal path between location A and B
- d) An accessible and user-friendly UI for VI by prototyping on PC for tractability and plan future phone porting

The core method for the QR code detection includes colour-selective detection under varying illumination, which can separate red/green/blue QR regions to shrink the search, geometric grouping to isolate candidate QR regions to validate the area that likely has a QR, and a vector-to-target computation from frame to the nearest reliable detection to guide the user on how to rotate and how close to move until decoding is stable. Next, the QR code reader module uses a standard library (OpenCV) to decode the QR code to a JSON file, which is a method that is widely used to provide a structural validation to verify that all required fields are present, their types and value ranges are correct, and they match corresponding nodes in the map. Additionally, the common method that is adopted for the route guidance module is path computation, where algorithms like A* and Dijkstra will be implemented here to compute the shortest path from the maps that have stored topology, corridors, and door nodes (database). Lastly, the mid-fidelity UI is finalized, and Flutter on Android will be used to design a fully functional UI.

The aims of the project are to create a cost effective, yet high-endurance end-to-end wayfinding baseline using purely visual information that is supplied by only printed QR codes and no physical site infrastructure. The offered approach proves that properly coloured QR markers, in combination with commodity mobile vision systems, are enough to localize, orient, and guide visually impaired users from using one room to another.

Area of Study (Maximum 2): **Computer Vision, QR decoding and error-correction**

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LIST OF ABBREVIATIONS

BLE	Bluetooth Low Energy
GPS	Global Positioning System
QR	Quick Response
VI	Visually Impaired
TTS	Text-to-Speed
CNN	Convolutional Neural Network
RGB	Red, Green, and Blue

CHAPTER 1 - INTRODUCTION

1.1 Problem Statement and Motivation

In recent years, technologies for people with visual impairments have advanced at an increasingly rapid pace thanks to great innovation and inclusive policies implemented by governments across the world. For example, people with visual impairments can navigate within indoor environments using beacon-based systems, which connect their smartphones to Bluetooth low energy beacons (BLE) to pinpoint their locations. Other technologies have also been used for the same purpose, such as smart glasses, artificial visual recognition, and so on. Nonetheless, these technologies come with several drawbacks, namely high battery consumption, high complexity, high costs and several others which can prove even more disadvantageous to those who are less privileged. Furthermore, in the case of Bluetooth based systems, devices with adequate Bluetooth capabilities are required for a stable and reliable experience, whereby beacons must also be installed throughout the building for the system to be fully realized. Technologies such as smart glasses and artificial visual recognition also require a strong and dependable internet connection to be functional, regardless, they are still in early development and can be unreliable at times.

This proposal aims to develop a robust and cost-effective indoor navigation system for the visually impaired using QR codes for route guidance. QR codes have seen wide adoption in Malaysia and across the world in the past few years and is incredibly easy to implement, it does not require extensive documentation or guidance as it is relatively accessible and easy to use for the average user. An indoor navigation system using QR codes alleviates the need to install any hardware in buildings, requiring only that QR codes be placed inside. It does not require a strong internet connection as it uses the global positioning system (GPS) to pinpoint the user's location and does not require additional hardware on the user's end where the user needs only to install an application to use the system. This system paves the way forward in the field of

technology for people with visual impairments and is a critical step in using technology to improve the lives of people with such disabilities.

1.2 Project Scope and Objectives

The scope of the project will be to develop an indoor navigation system for the visually impaired using QR codes, which will include four modules namely a QR code detection module, a QR code reader module, a route guidance module, and a user interface. The QR code that will be used for the purposes of this project will be QR code model 2, whereby other types of QR codes such as micro QR codes or secure QR codes will not be considered. The project will incorporate the four modules into one single robust system, where the user interface will be primarily developed with accessibility features in mind for the visually impaired with less of a priority on visual design. Due to time and budget constraints, the QR code reader will only be designed to detect a handful of colours. The user interface will also have limited audio recognition and voice feedback capabilities, with the focus being system accuracy and reliability to deliver a fully functional product.

The project aims to deliver a functional indoor navigation solution for the visually impaired that will help them navigate accurately within indoor environments. To evaluate the performance of the system, the following benchmarks were used:

- QR code detection rate: The QR code detection software should have a success rate of 95% and above when used within the set parameters
- Location accuracy: The location accuracy of the QR code should not exceed the maximum margin of error of 2 meters
- Audio recognition: The system should have a speech-to-text accuracy of 90% of above by measuring the number of words detected accurately
- Voice feedback: The system should be able to give accurate route guidance instructions with a 100% success rate given that it has received the correct voice input

1.3 Impact, Significance and Contribution

Many systems have been implemented throughout the years to improve indoor navigation for the visually impaired, but very few exceed the benefits of a QR-based indoor navigation system. Firstly, because a QR-based system does not require significant upfront costs, building owners will be more incentivized to implement such a system as the estimated return-on-investment is much higher compared to other systems. Besides that, QR-based systems have a higher probability of being widely adopted by the market since they are easier to implement compared to other systems. If these impacts are realized, people with visual impairments will be able to navigate indoors safely in various buildings without relying on expensive hardware, which can be burdening for some. Visually impaired folks will also be able to navigate around large buildings such as shopping complexes or hospitals with less effort, opening new possibilities which would otherwise have been unsafe or too inconvenient.

As technology becomes increasingly integral in our everyday lives, there is a need to ensure accessibility and inclusivity for those who are less privileged in our society. The implementation of QR-based indoor navigation systems hopefully addresses such needs, while raising the awareness of other users regarding the needs and the inconveniences of the visually impaired. In the long term, the goal of this project is not only to improve the lives of folks with visual impairments, but to hopefully cultivate empathy and mutual understanding among users from all walks of life. In conclusion, this project plays a pivotal role in promoting inclusivity in the use of technology while contributing greatly to the field of technology for the visually impaired and other disabled folks.

1.4 Background Information

An indoor navigation system is a system that provides route guidance to the user within indoor environments. This is achieved by using one or more of the following wireless technologies such as GPS which uses multiple satellites orbiting around medium earth orbit to pinpoint the

user's location, Wi-Fi which uses real-time Wi-Fi signals emitted by the user's device to determine their location, Bluetooth which is used to determine the angle of arrival or departure of a connected device to triangulate its location with sub-meter accuracy, and cellular data which can approximate the user's device location based on the location of the connected cell tower. A mapping of the indoor environment is stored in a remote database, which is used to determine the user's location relative to the map. The system is then fully realized through the implementation of software typically including features such as step-by-step voice navigation, location searching, offline capabilities and so on. The implemented software should be simple and user-friendly to ensure a good user experience.

A QR code is a type of two-dimensional code that stores information in a square grid vertically and horizontally. A QR code can encode various types of information such as text, images, audio, videos and so on. It can also be used for retrieving or submitting information for various purposes such as attendance tracking, online payments and so on. The QR code can be activated by being scanned by any devices with a QR code scanner, it can be scanned in any orientation as it has three large squares in the corner to help the scanner determine the QR code's direction. QR codes have built-in error detection, meaning that it can still be scanned if a portion of the QR code has been damaged.

Technology for the disabled encompasses any technology that is designed with accessibility in mind for people with disabilities such as hearing difficulties, visual impairments, mobility issues and so on. These technologies can include but are not limited to hearing aids, braille screen readers and electric wheelchairs. However, common technological devices such as smartphones, smartwatches and augmented reality devices can also include accessibility features to enable people with disabilities to use them. These devices typically include features like text-to-speech, screen magnifiers, speech recognition and many more that can assist the disabled in using devices that were not primarily marketed towards the disabled. The use of

QR codes for indoor navigation for the disabled is an example of such a case where a technology that was not necessarily meant for the visually impaired can be used to improve accessibility for the visually impaired.

A user interface is the point of communication and interaction between the user and the system, which includes interactive elements each representing a feature of the system. These elements allow the user to convey information to the system while also simultaneously receiving information from the system. A user interface in its most common form includes a screen that can be interacted with either through a device or by touch, although there are also other user interfaces such as voice user interfaces, command line interfaces and so on. By design, a user interface should be user-friendly and easy to use, it should also be responsive and intuitive enough to allow for a good user experience.

CHAPTER 2 - LITERATURE REVIEWS

2.1 Previous Work on QR Code Detection

QR code detection has become a crucial tool in modern technology, offering convenience and efficiency in everyday tasks. From making payments and accessing information to applications aimed at assisting visually impaired (VI) individuals with indoor navigation, QR code detection is used. According to V. Jain et al., there are three key stages in obtaining information from a QR code in real-world environments, namely localization, image pre-processing, and decoding, as shown below in Figure 2.1.1. First, localization refers to identifying the presence and exact position of the QR code within an image. Next, image pre-processing serves to enhance the quality of the detected QR code by reducing issues such as blurred QR codes, distortion, angular perspective, etc., thereby improving the likelihood of successful decoding. Finally, decoding extracts the encoded information, where it relies on the main standard architecture of the QR code [9].

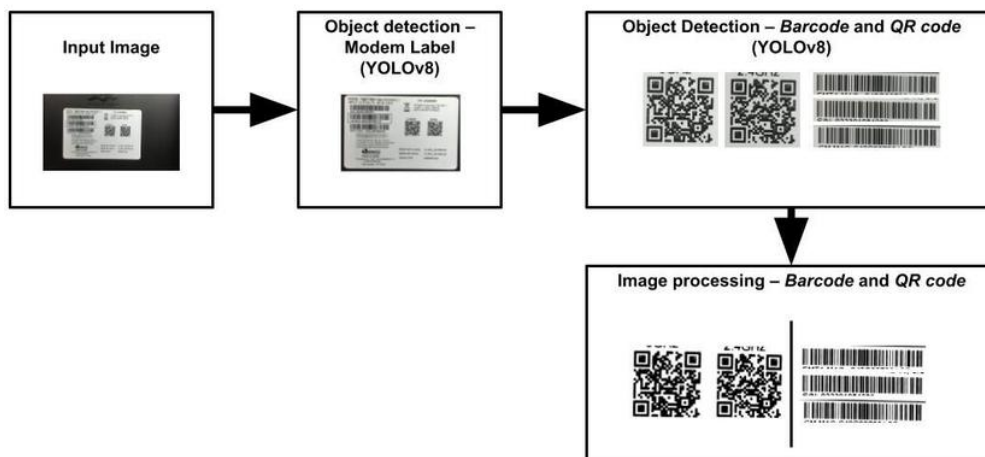


Figure 2.1.1 Workflow of QR Code Detection: Localization, Preprocessing, Decoding [13]

Recent studies show a growing use of deep learning techniques, such as YOLO and SSD, which offer high accuracy in detecting QR codes [9]. As technologies advance with mobile devices becoming more powerful, the focus has shifted from speed to accuracy. To improve accuracy, researchers have used Convolutional Neural Networks (CNNs) to widen the detection angle, making it easier for VI users to scan codes without precise camera alignment. Furthermore, W. C. Kurniawan et al. stated that hybrid methods that combine traditional-based methods like Pyzbar with deep learning are being explored to improve both speed and robustness in real-world scenarios [10]. The comparisons of different deep learning techniques, which are YOLO, SSD, and the hybrid approach, can be viewed in Table 2.1.1 as shown below.

Method	Strengths	Weaknesses	Suitability
YOLO	• High detection accuracy • Real-time processing speed	• Requires powerful hardware for peak performance • Performance may drop at the point of poor lighting	Excellent for real-time detection and guidance, especially in mobile devices with sufficient GPU/CPU power
SSD	• Fast and accurate for medium to large objects • Can be optimized to require fewer resources than YOLO, enabling faster performance on lower-powered devices • Good trade-off between speed and accuracy	• Slightly less accurate than YOLO for small/angled QR codes • May require tuning for low-light conditions	Suitable for applications prioritizing speed over highest possible accuracy
Hybrid (Traditional + Deep Learning) (e.g., Pyzbar + CNN)	• Combines strength of both methods • Better detection across wide angles • Can reduce false negatives in challenging environments	• Integration complexity • May be slower if not optimized	Ideal for VI navigation where robustness and tolerance to imperfect alignment are crucial, even at slight cost to speed

Table 2.1.1 Comparison of YOLO, SSD, and Hybrid Methods for QR Code Detection for QR Code Detection in Indoor Navigation for VI users

Despite great advancements, several challenges still exist. Since findings frequently differ between systems and datasets, one significant drawback is the lack of a consistent benchmark dataset, which makes it challenging to evaluate approaches fairly [9]. There are further complexities that real-world environments introduce, including complex backgrounds, perspective distortions, and the need for precise camera framing [10]. Moreover, M. L. Hakim et al. observed that detection performance is affected by distance and angle, as vertical scanning ranges are shorter due to camera field of view (FoV) constraints, which is crucial for VI users who may scan from different heights [11]. Lastly, low-resolution images and poor lighting conditions still pose problems, highlighting the need for more robust and adaptable detection techniques [9].

Thus, in order to achieve fair performance evaluation, future research should concentrate on comparison testing of detection algorithms using consistent hardware and datasets. Some examples include YOLO, SSD, and multi-scale pyramid pooling [9]. Furthermore, exploring hybrid approaches that combine conventional and deep learning methods may offer faster and more robust detection, especially for wide-angle scanning [10]. Last but not least, further work is needed to improve detection in low-light conditions and to develop techniques capable of handling severe image blurriness in actual indoor settings [9].

Aspect	Key Points	Strengths	Weaknesses
Core Process	Localization → Image Pre-processing → Decoding	Structured 3-step process ensures systematic detection and reading	Requires precise execution at each step for reliable results
Deep Learning Methods	YOLO, SSD	High accuracy and robust to certain variation	May require higher computational power
CNN Angle Widening	CNNs increase detectable angle	Easier scanning dependent on lighting and camera quality	Still dependent on lighting and camera quality
Hybrid Approaches	Traditional (e.g., Pyzbar) + Deep Learning	Improved speed and robustness	Integration complexity and may need more tuning
Distance & Angle Effects	Vertical FoV smaller than horizontal which affects scanning from different heights	Recognizes the importance of camera orientation	Longer detection times at greater distances

Table 2.1.2 Summary of QR code detection Research for Indoor Navigation for the VI

Table 2.1.2 provides a consolidated view of key approaches and factors influencing QR code detection, highlighting how methods such as deep learning, CNN-based angle widening, and hybrid approaches contribute to improved accuracy and usability. From this comparison, it proves that while these techniques have advanced detection capabilities, achieving reliable performance in varied real-world conditions will require addressing the remaining issues, such as inconsistent datasets, lighting variations, and distance-related detection challenges.

2.2 Previous Work on QR Code Readers

The QR code reader is used to decode QR code markers to determine the user's current position within a map. According to A.Idrees et al., the common or the current trend in technology implementation for this particular module is that each QR code marker represents a unique node with encoded information such as its ID, name, coordinates, and possible nearby movement directions (left, right, forward, backward, and number of steps). The system retrieves contextual information from the database and informs the user about their current location using a text-to-speech module upon detection. This implementation of QR code reader is not only able to recognize the user's position but also initiates navigation through accurate audio feedback, without requiring an internet connection [5]. Some solutions also incorporate monocular depth estimation for obstacle avoidance. [4] and mobile trend is combining QR positioning with graph search and AR overlays to provide intuitive steering. [14]

Research	Method Proposed regarding QR code reader module
[5] A.Idrees et al. 2015	• QR strips laid on the corridor floor • QR code reading by ZXing → query local SQLite map library → TTS broadcast • Support Deviation Detection for yaw correction (roll back to the last correct node).
[14] Y.Alkady et al. 2024	• ARCore+SLAM on mobile • QR code to set location provide grid map • Support Theta* path planning • AR overlay arrow navigation
[4] A. Fraga et al. 2022	• QR serves as a positioning beacon on the wall • The entry QR downloads the entire building database and then goes offline • Support monocular depth obstacle avoidance.

Table 2.2.1 Summary of method proposed for QR code reader module for each research

To be convenient for the VI user, A.Idrees et al. stated that the indoor navigation is achieved using Android smartphones and QR code strips on the ground, as shown in figure 2.2.1 below. The step-by-step process involves scanning the code to read the current location by using ZXing, which is suitable for scanning large-sized images. The QR code instructions will then be fetched from the SQLite database, which carries the information about all nodes of a particular scenario, and provide output (current location) on the screen. After that, it will proceed with the common process of the navigation system to generate path planning, and TTS announces the next QR code. Moreover, if the user deviates (goes in the wrong direction), the system also adapts with Deviation Detection, which guides the user back to the last correct node and continues onward. Overall, the code reading and announcement are completed entirely on the mobile phone, emphasizing an accessible interface without additional hardware and network dependence, and as a result, in the tests, it can still be recognized in low light and up to 60% blur. [5]

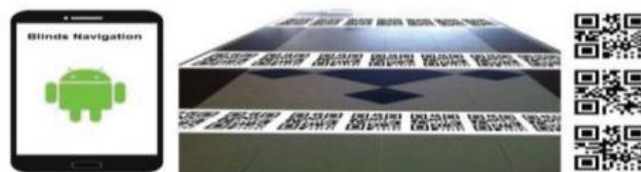


Figure 2.2.1 Floor QR strips method [5]

Furthermore, Y.Alkady et al. highlighted an innovative method that utilizes mobile ARCore/SLAM as its core, this system enables GPS-free indoor positioning and navigation.

The phone's camera and sensors track the user's position in real time, serving as the foundation for AR overlays. At this point, the QR code reader module will discretize the indoor environment into a grid map, and the user will scan a QR code to set or correct their current location. After that, arrows, road signs, and turn instructions are then overlaid on the real-time image. For the part of path planning, it utilizes Theta*, which, compared to A*, allows for diagonal connectivity within the line of sight and reduces unnecessary turns through "theta pruning" to guide users along a smoother and more natural path. When the user scans a new QR code or changes their destination, the system dynamically recalculates the path and updates the AR navigation instructions. In summary, the paper defines the overall pipeline as comprising a mobile application, a grid system, QR code scanning, AR overlay, path planning/navigation, user interface and instructions, and location-based services, emphasizing integrated implementation on the phone. [14]

Moreover, A. Fraga et al. [4] proposed a cost-effective and feasible method for almost any environment where QR codes are used as location beacons and linked to an online database, and the code contains a node ID, coordinates, adjacent nodes, and actions. Upon entry, the user downloads the entire building database from the entrance QR code and is then guided offline. The system provides two modes: "Normal Mode" reads the code and announces the current location and context while "Automatic Mode" calculates the shortest path and recalculates the path when an unexpected QR is recognized. With the illustration by A. Fraga et al. [4], the system demonstrates that each QR code marker, strategically placed at key indoor locations, functions as a unique identifier containing location and directional information. The QR code reader captures this information and, by referencing a preloaded database, enables the system to calculate the optimal path to the user's chosen destination. This process facilitates both real-time location tracking and path planning, which are crucial for implementing a QR code-based navigation module that provides guidance based on encoded directional data. For example,

when the initial point is marker 7 and the destination is marker 3, the system utilizes the path calculation algorithm and the QR code database to determine the optimal route, which, as shown in Figure 2.2.2, follows the sequence (8, 4, 2, 3).

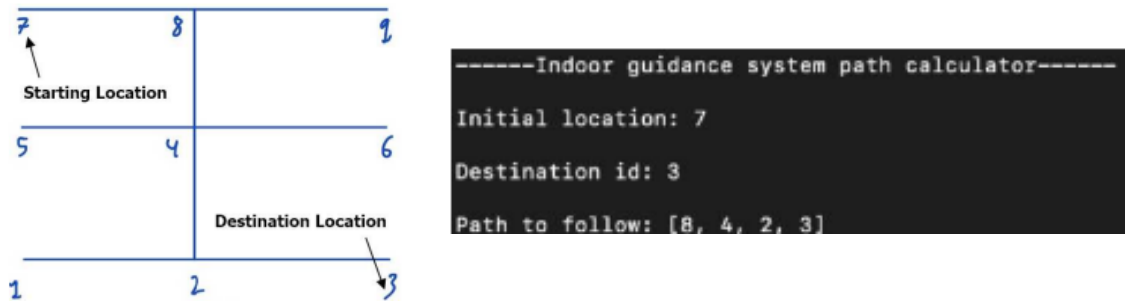


Figure 2.2.2 Path calculation algorithm that adopt by QR code-based navigation [4]

Additionally, the system also includes obstacle avoidance that is achieved by combining monocular depth estimation, and all commands are output via TTS. Monocular depth estimation is used to calculate the distance from each pixel in front to the camera from each RGB image frame. A "safety corridor" is drawn in the planned direction of travel. If the closest distance within this corridor falls below a threshold, an obstacle avoidance strategy (stop, sideways movement, or detour) is triggered. All commands are outputted to the user via text-to-speech (TTS). [4] Figure 2.3.3 shows obstacle avoidance detection of an obstacle at a safe distance (left) and obstacle avoidance detection of an obstacle about to hit a person (right).

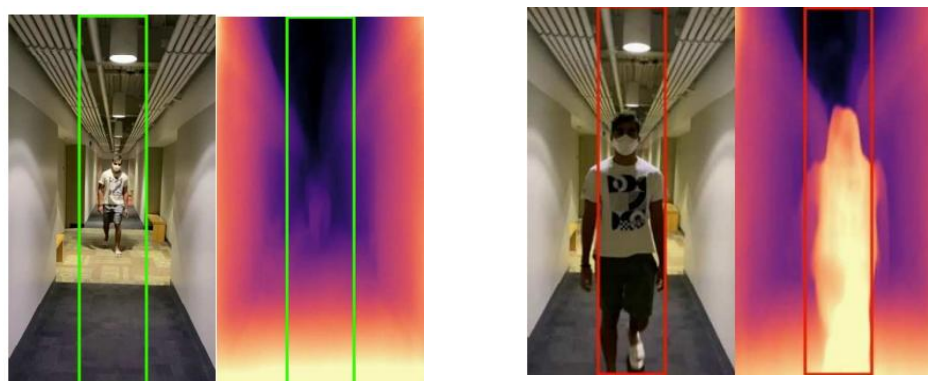


Figure 2.2.3 obstacle avoidance detection

To adopt one of these methods, below are the advantages and limitations of each method, allowing us to select the most achievable and cost-effective module to integrate into our system.

Method	Advantages	Limitation	Suitability
Floor QR strips-method	• Low cost • Available offline • Readable even in low illumination and up to 60% blur • Mature gesture interaction designed for the blind • Completed entirely on the mobile phone	• The app needs to be kept running with the camera on to avoid missing scans • Need to place large amount of QR codes on the floor in strips across the corridor, approximately every four steps	• Flat ground • Well-protected ground • Single route • Places with high requirements for wall protection
AR-QR overlay method	• The experience is intuitive, with clear steering alignment • The path is smoother and there are fewer turns • Fully compatible with the "mobile terminal"	• Implementation is complex • Requires high computing power • Theta* does not guarantee global optimality	• Needs an ARCore compatible device • Emphasize immersive experiences
QR-beacon method	• VI-friendly • Runs offline • Obstacle avoidance for greater safety • Can be ported to mobile • Phones/low-power accelerators	• Requires early database construction and code distribution • Mainly PC/camera prototypes, with fewer details on the mobile side.	• Locations with weak network signals • Locations that can be placed on doorways, corners, or walls.

Table 2.2.2 Comparison of Floor QR strips, AR-QR overlay and QR-beacon for QR Code Reader

In summary, [5] focuses on the end-side reader: using ZXing decoding, taking the node_id to the local SQLite to fetch instructions, and then broadcasting them through TTS. Its advantages are offline, robustness, and yaw/fallback correction. In the paper "Using QR as a positioning beacon": the code/library carries structured information such as node_id/coords/neighbors/action. The entry code downloads the entire database at once and then runs offline [4]. Its advantages are "read code, action" and the code can be recalculated in case of an unexpected error. [14] With AR visual guidance as the core, QR only serves as a location anchor, and the grid + Theta* provides a smoother path with fewer turns. Its advantage is intuitive steering prompts.

The main gaps shared by the three are insufficient reader implementation details. Many papers emphasize "scan the code to navigate" but do not explain how to scan, what to scan, and what to do if it is wrong in terms of library selection, camera parameters, payload verification, failure

recovery, and latency indicators. If it is wrong on-site, it will be impossible to read, read incorrectly, or read too slowly, and it will be even worse when errors appear.

Based on this, our reader selection is as follows: adopting the structured payload (node_id/coords/neighbors/action) of [4], combining the offline operation and yaw/retreat correction of [5], and introducing the direction guidance of [14] on the UI side (simple voice guidance). The process is Scan, Decode, QR code Structure verification, Check library/update, announce "turn/forward" and Go to the next QR or unexpected code to trigger recalculation. For example, scanning the code to obtain the payload {"node_id":"N-012","neighbors":["N-013"],"action":"R45","coords":[12.3,5.8]}, the system will immediately announce "turn right 45 degrees, go straight 7 meters to N-013" after verification; if it mistakenly enters and scans the unexpected node N-021, it will immediately recalculate and change the broadcast to "turn left 30 degrees, return 2 meters to the direction of N-013". This solution complements Reader's feasibility and on-site robustness while maintaining low cost and offline availability and leaves a path for mobile phone and AR upgrades.

2.3 Previous Work on Route guidance

Modern developments in assistive navigation technologies for visually impaired individuals (VIPs) show a progressive transition from conventional path-planning methods toward intelligent systems that integrate real-time perception, personalized guidance, and adaptive mobility. Nawaz et al. [6] offer a foundational review of navigation algorithms commonly applied in assistive systems, such as A*, Dijkstra, and various artificial intelligence models, including Particle Swarm Optimization. While these techniques offer computational efficiency for static route generation, the study pushes out critical restrictions in real-world application, especially the lack of contextual awareness, limited adaptability, and an inability to respond dynamically to unpredictable environments. Expanding upon this, Saparudin et al. [7] presented a GPS- and GIS-based navigation assistance that delivers waypoint-driven voice

guidance and provides practical benefits for outdoor mobility. However, the system's dependence on static map data and the absence of real-time obstacle detection mechanisms diminishes its effectiveness in complex, dynamic settings where time-to-time situational feedback is essential.

To address these weaknesses, Chan and Tan [8] present *SIGHT*, a vision-based assistive system that leverages lightweight deep learning models—YOLOv8 for object detection and MiDaS for depth estimation—implemented totally on smartphones. This approach enables real-time environmental sensitivity without the need for external hardware and offers both scalability and enhanced user autonomy. By incorporating live hinder detection and spatial awareness into a mobile-friendly platform, *SIGHT* represents a significant advancement toward practical, real-world navigation support for VIPs. Together with all these investigations, it had illustrated an evolution in assistive technology, from rule-based routing and static map systems to sensor-fused, AI-powered navigation tools, which support interactive, context-aware mobility.

For all these advancements, an obvious research shortage remains in the development of unified, end-to-end navigation systems that integrate global path planning, local hazard recognition, environmental context, and personalized user feedback into a single, cohesive solution. The summary of these researches highlights the need for integrative innovation and creativity—connecting computer vision, geographic information systems, artificial intelligence, and human-computer interaction—to hand over safe, intelligent, and self-operating navigation experiences for visually impaired users across diverse settings.

Research	Approach	Strengths	Limitations
Nawaz et al.	Review of navigation algorithms (A*, Dijkstra, AI models incl. Particle Swarm Optimization)	• Efficient for static route generation • Foundational understanding of assistive navigation methods	• No contextual awareness • Limited adaptability • Cannot respond to unpredictable environments
Saparudin et al.	GPS- and GIS-based navigation with waypoint-driven voice guidance	• Practical for outdoor mobility • Voice feedback improves usability	• Relies on static map data • No real-time obstacle detection • Less effective in dynamic settings
Chan & Tan - SIGHT	Vision-based system YOLOv8 (object detection) + MiDaS (depth estimation) on smartphones	• Real-time environmental awareness • No external hardware required • Scalable and portable • Supports user autonomy	• Focused on local hazard detection • No full global path planning integration

Table 2.3.1 Summary algorithmic routing to GPS voice guidance and real-time vision in Indoor Navigation for VI users

2.4 Previous Work on User Interface for the Visually Impaired

The user interface of an indoor navigation system for the visually impaired using QR codes is designed for ease of use and accessibility. This includes key features such as navigation guidance, current location information, and destination selection. Existing systems also incorporate user-friendly gestures like single tap for repeating instructions, double tap for confirmation, and swipe gestures for navigating through different pages. Features such as audio feedback are also included with accurate speech recognition models to enhance the functionality of the system. [1] Besides this, some other systems that are also designed for visually impaired users leverage the Android system to take advantage of its existing features. These systems sometimes prioritize using voice commands to navigate throughout the user interface instead of incorporating gestures to increase user-friendliness and minimize user training. [2]

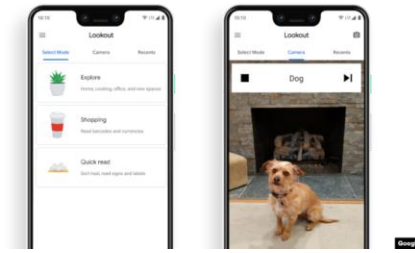


Figure 2.4.1 *A user interface of the Lookout app designed by Google that includes navigation and image recognition capabilities for the visually impaired [3]*

Current research elaborates on the inclusion of accessibility features such as text-to-speech, voice feedback, and swipe gestures that assist the visually impaired. However, these studies rarely mention in detail the implementation of such accessibility features. Some of the problems that may arise from the user interface could include information overload, unclear audio presentation, lack of tactile feedback, and so on. If the elements of the user interface are presented all at once in a sequential manner, the user might be overloaded with too much information and will be unable to navigate the user interface. Moreover, the user interface might come across as unintuitive without any tactile feedback given to the user. In the application of an indoor navigation system for the visually impaired, tactile feedback could increase the accuracy of user inputs by providing immediate physical responses so that users can confirm that their inputs have been registered correctly. Feedback confirmation through tactile feedback has also been proven effective in critical applications such as those used in the automotive industry and in medical devices. In fact, existing research has shown that tactile feedback and audio cues are not only essential to a good user experience but can also lead to a safer experience. [12]



Figure 2.4.2 Haptic feedback on a mobile touchscreen which mimics tactile feedback to provide confirmation cues to users

To improve these weaknesses, further research is required to ensure that the implementation of the user interface is defined in a more detailed manner that is user-friendly and intuitive for the visually impaired.

CHAPTER 3 – PROPOSED METHODS

3.1 Design Specification

This project created a mobile application that utilizes QR Code Model 2 to carry out navigation tasks for visually impaired individuals in indoor environments. QR Code Model 2 is the commonly used model today which has an alignment pattern to increase the robustness and readability of the QR codes. The system acquires the QR image through the device camera, and sends it to a Python-based browser, which uses libraries such as OpenCV to decode the QR code. The location data is processed by a route guidance module that sends audio-based navigation instructions. The mobile application runs directly into camera mode to simplify interaction complications and allows basic voice commands, enabling hands-free operation. The system architecture is made up of three main components: the mobile user interface, the Python server-side, and the data transfer layer. The mobile front-end is responsible for capture images, playback navigation guidance, and voice command interpretation. The backend processes these images, decodes QR content, and generates navigation instructions. The data transfer layer is responsible for the secured transfer of the data between the user and server.

3.1.1 Requirements

Functional requirements comprise detecting and decoding (scanning) QR codes, providing real-time route guidance, generate clearly defined audio navigation instruction, permitting voice commands, and leaving open functionality. The application being created is available for Android and iOS devices and can scan QR codes in under five seconds of an application launch and allows for good fast server-side communication to minimize delays. Non-functional requirements focus on performance and reliability. This system needs to achieve a QR code detection performance of 95% and a speech to text performance of 90% accuracy for commands. Audio output must be clean with no extraneous noise. The backend must produce navigation instructions in under two seconds with all image data and location

data being secured by HTTPS with the intention of protecting the user's privacy. The application has been created for long-term stability to the applicable accessibility standards which allows a visually impaired user to use this application by voice and without near vision interaction.

3.1.2 Software

The mobile app will be built using a cross-platform development framework, such as Flutter, to ensure compatibility with both Android and iOS, and the backend will be implemented in Python 3.10+. QR code processing will be based on OpenCV for image handling and Pyzbar for QR decoding. Voice command input will be processed via Google Speech-to-Text or other similar engines. Any audio navigation output can be processed using text-to-speech engines supplied by Google TTS or other similar products. Communication between the frontend and backend will be secured by REST APIs, based on HTTPS calls. UI/UX design will be generated using assistive design processes. A rough design will be created using Adobe XD. Testing and development will utilize Firebase Test Lab on Android devices, TestFlight on iOS devices, and Git-based version control system using GitHub.

3.1.3 Hardware

Computer Requirement	
Component	Minimum Specification
CPU	Intel Core i5 (10th Gen) or AMD Ryzen 5 equivalent
RAM	8GB
Storage	256GB SSD
GPU	Optional
OS	Windows 10/11, macOS 12+
Network	Stable broadband connection

Table 3.1.1 Minimum hardware specifications for computer

Mobile Device	
Component	Minimum Specification
CPU	Quad-core ARM processor (e.g., Snapdragon 665 or Apple A10)
RAM	3GB
Storage	64GB
Camera	8 MP with autofocus
Microphone	Built-in
OS	Android 9.0+ / iOS 13+
Connectivity	Wi-Fi or 4G/5G

Table 2.1.2 Minimum hardware specifications for mobile devices

3.1.4 Data Specification

Data specifications include both metadata and image data. Metadata will store the timestamp of each scan, device ID, decoded location ID, command type, navigation status, and lighting conditions at the time of capture. Image data will be stored in JPEG or PNG format with a minimum resolution of 640×480 pixels and compressed to a maximum size of 200 KB per image. Images will use the RGB color space. This configuration ensures efficient data transmission while maintaining accurate QR code decoding.

3.2 System Design / Overview

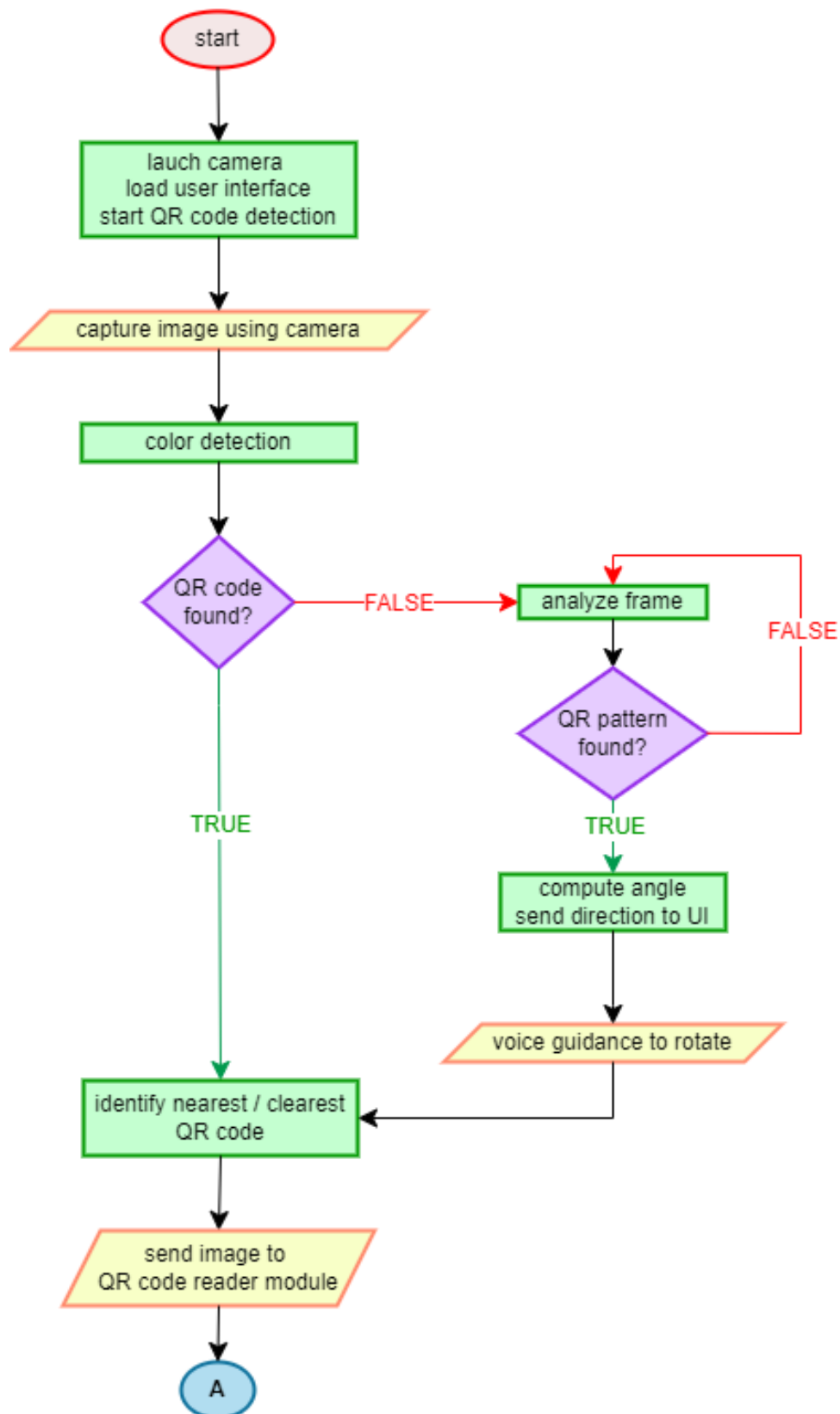


Figure 3.2.1 Flow Chart of Stage 1 - QR code detection

To simplify the system design of our indoor navigation system, three stages would be proposed and illustrated in this covered part. Figure 3.2.1 shows the first stage about the QR code detection module. When the user launches the Indoor Navigation System, the application will automatically initialize the camera module and the user interface (UI). Once initialized, the system will immediately activate the QR Code Detection Module to begin scanning.

The detection process begins by capturing an image using a camera as input. The system uses image processing techniques to detect coloured QR codes, such as red, green, or blue. Using colored QR code markers allows the system to quickly narrow down the potential detection area within the frame, reducing computational load and false positives. By focusing only on areas that match the expected color pattern, the system can detect valid QR codes more efficiently and accurately. When the system detects an area that might be coloured, it then determines whether a valid QR code is present there. The system transmits the captured image to the QR code reader module for additional decoding after choosing the closest and clearest QR code marker if a QR code detection is successful. If no QR code is found, the system will analyze the current frame to detect possible QR-like patterns based on colour and shape. Although a QR code is not yet fully recognized, this preliminary analysis allows the system to compute a vector from the center of the screen to the possible QR code direction based on size, position, and colour match. The system determines the necessary rotation angle (e.g., "rotate right 30°") using this vector information, then transmits the result to the user interface. The user interface then helps the visually impaired user position the camera toward the QR code by providing on-screen visual cues and audio-based speech assistance output (e.g., "Please rotate slightly to the left to detect the QR code"). This loop continues until the QR code is centered and fully captured, allowing the next module to take over.

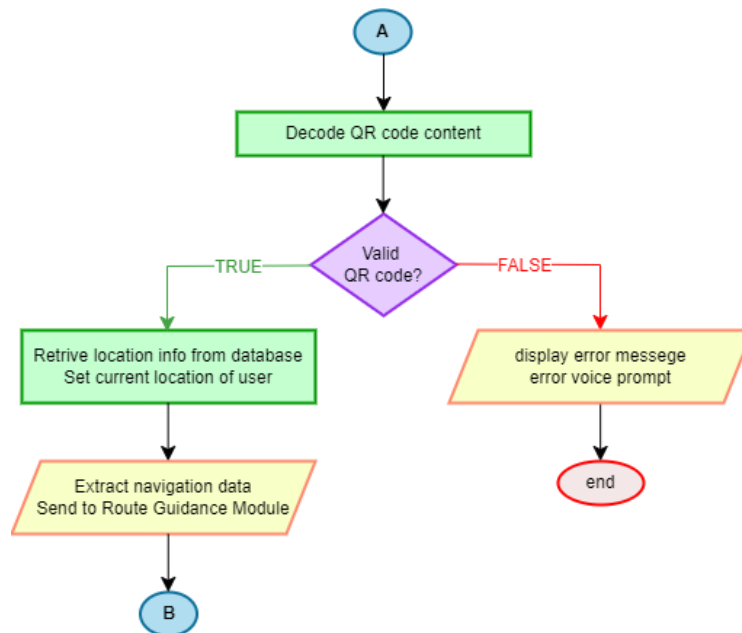


Figure 3.2.2 Flow Chart of Stage 2 - QR code reader

According to Figure 3.2.2, move on to stage 2. When the QR code image is passed from the QR code detection module to the QR code reading module, the system first performs a system QR code verification process to verify that the QR code contains the correct system ID or signature. If validation fails, the user interface (UI) displays a warning stating, "This QR code does not belong to this system," and plays a voice prompt, "A QR code outside the system has been detected. Navigation has been terminated.", and immediately terminates execution and does not continue route calculation. In contrast, the system provides a location ID (uniquely identifying the current location) and direction data (including predefined navigation instructions stored in the system database, such as "Turn right 45 degrees") and sets the user's current location based on the location ID and finally forwards the extracted data to the route guidance module for further processing and calculation of the navigation path.

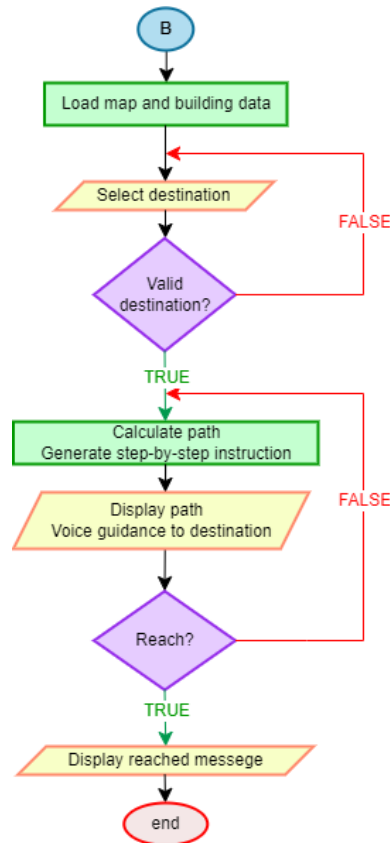


Figure 3.2.3 Flow Chart of Stage 3 – Route Guidance

According to Figure 3.2.3, Stage 3 corresponds to the route guidance module. This module first loads a map structure for the current floor or building (e.g., the floor plan of Building N) from the system database. The map contains the location (location ID) of each QR code marker and its adjacent nodes (i.e., possible connection points). The user selects the desired destination through voice input on the user interface. After the user selects the destination, the system first verifies the clarity and correctness of the input. If the voice destination is unclear, the system prompts the user to repeat it. Once the destination matches a valid entry in the system database (e.g., "N104"), the route guidance module uses a path planning algorithm to calculate the optimal route—whether based on shortest distance, minimum number of steps, or minimum turn angles. Once computed, the system generates step-by-step navigation instructions (e.g., “Turn right 45°, walk 5 steps” or “Turn left 90°”) with an estimated arrival time. These instructions are sent to the UI for graphical display of the path and output via voice guidance.

At each node, the system provides the next instruction until the user reaches the destination. The module continuously checks whether the destination is reached. If reached, the system terminates navigation; otherwise, it keeps outputting updated step-by-step instructions until arrival.

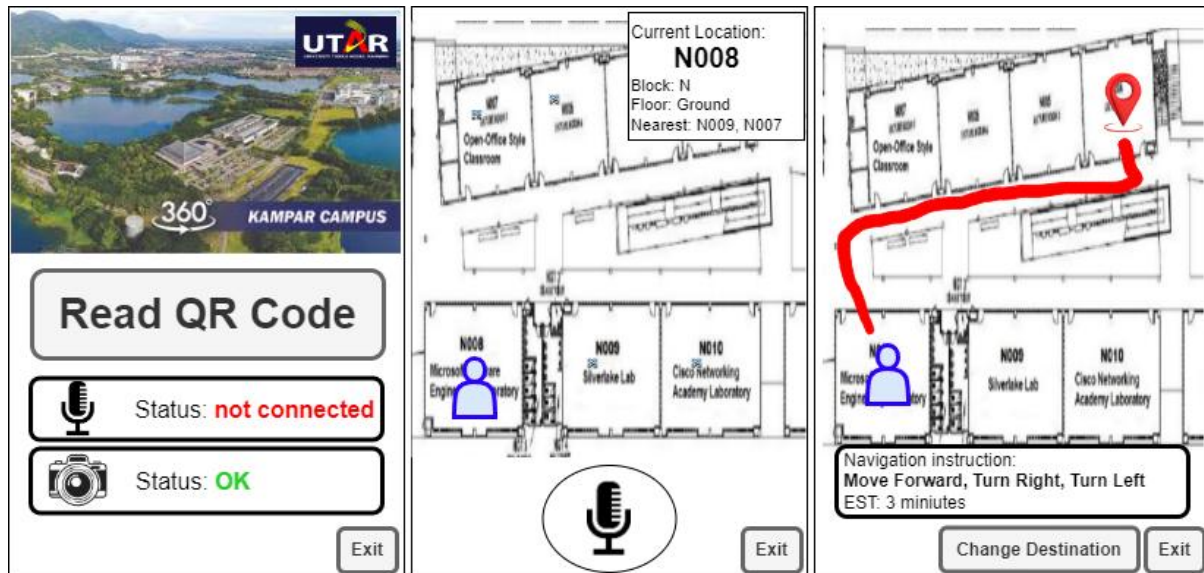


Figure 3.2.4 User Interface for System Home page, Map page and Navigation Mode page

Based on Figure 3.2.4, the interface design is optimized for visually impaired (VI) users with a clear, minimal, and intuitive layout to reduce interaction complexity and cognitive load. The interface consists of three main sections: Homepage, Map Page, and Navigation Mode Page. The homepage's header image showcases a system usage scenario, using Universiti Tunku Abdul Rahman (UTAR) as an example. A "Read QR Code" button is located in the centre for quick access. Pressing this button launches the QR code scanning interface. Successful scans direct the user to the map page. The homepage also includes a status check for audio and camera connections, with a green indicator indicating a successful connection and a red indicator indicating a connection failure.

The map page displays the user's current location on a map with a blue icon and a details panel showing the block, floor, and nearest location. A microphone icon at the bottom allows users to search for destinations using voice commands, enhancing accessibility for VI users.

Once a destination is determined, the system marks it with a red pin and generates a red route to guide the user. In navigation mode, the system displays step-by-step text-based instructions, while voice navigation provides the same information and an estimated time of arrival. Upon arrival, the system announces, "Navigation successfully completed" and automatically returns to the homepage, prompting the user to scan a new QR code to update the next route.

3.3 Implementation Issues and Challenges

Implementing a cost-effective indoor navigation system for the VI using QR codes presents several challenges. One of the most significant challenges is detecting QR codes in real-world conditions. For instance, there may be times where lighting varies, sticking QR codes on glass doors may cause reflections, and long-distance scanning can reduce accuracy. Further complexity is introduced when detecting QR codes in different colours, such as red, green, and blue, because colour identification needs to be resistant to background interference and lighting variations. Moreover, guiding users to a QR code requires accurate vector computation and responsive feedback, which can be difficult to provide while the user is moving or not aligned with the field of view of the camera. To accurately read the QR code, the reader must interpret the code correctly even under distortions, motion blur, or partial obstruction, then map the location to a stored floor plan. The system's route computation, which determines the best route between points A and B, must smoothly integrate with this mapping process. In order to ensure that the navigation instructions are both comprehensible and useful in real-world situations, the system must not only compute an accurate route but also present them in a way that is easy for the VI user to follow. For example, include clear, timely audio cues or, during testing, an intuitive PC-based display for verification.

The proposed system integrates dynamic route guidance, robust colour recognition, and real-time QR code detection into a single, cohesive VI navigation solution. This design uses directional guide vectors to dynamically lead the user towards the closest code, in contrast to

traditional QR code systems that are frequently static and require exact positioning. In order to increase flexibility without requiring additional infrastructure, the method also investigates using color-coded QR markers to depict several navigation routes on a single floor map. Finally, the system's modular design, which includes QR detection, QR reading, route computation, and a VI-oriented user interface guarantees that each component can be updated or modified for future smartphone deployments with little change.

3.4 Timeline

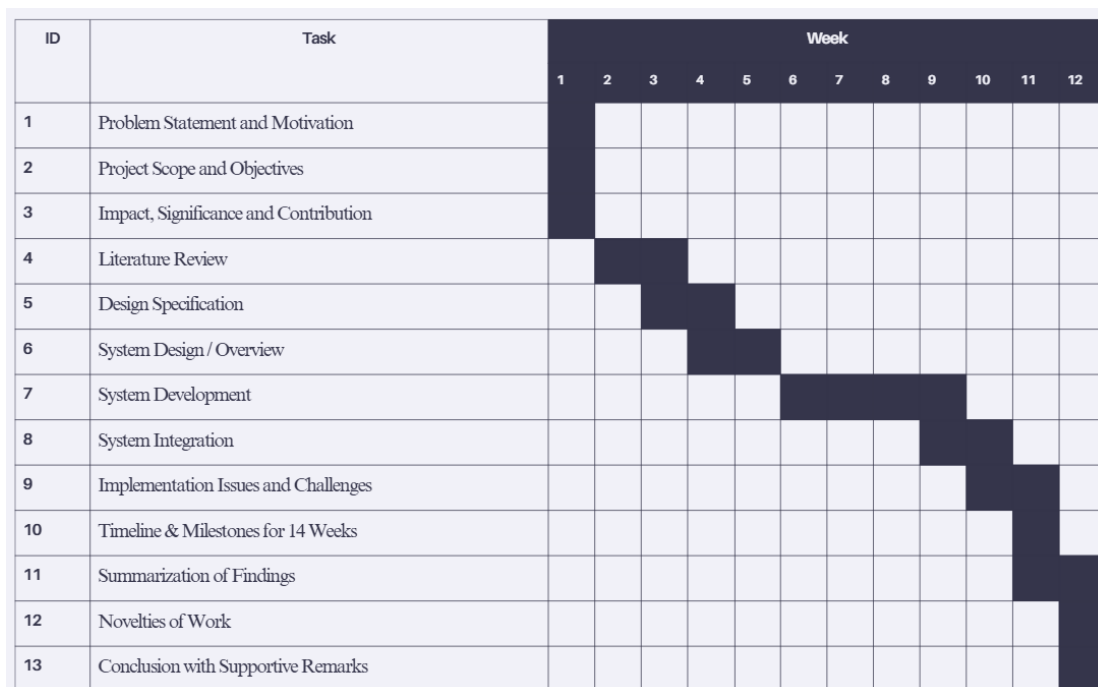


Figure 3.4.1 Project Gantt Chart

The 12-week project timeline has been structured to allow the development process to be completed within a reasonable timeframe and in a logical manner. It begins in week 1 with the formulation of the problem statement and motivation as well as the project scope and objectives, which establishes a clear foundation for the project. This is followed by identifying the impact, significance, and contribution of the project and conducting a thorough literature review from weeks 2 to 3 to understand existing work and justify the project's relevance. Two weeks have been allocated for the writing of the literature review to ensure that there will be no overlap with previous research in this study. In weeks 3 to 4, the design specifications are laid out,

which then leads into the system design or overview phase in week 5 and 6. The system development phase has been scheduled to be completed from week 6 to 9, where the four modules, namely the QR code detection module, QR code reader module, route guidance module, and user interface module, are implemented. Once development is completed, the modules will be integrated into a complete system from week 9 to 10, ensuring that different modules work together seamlessly. Weeks 10 to 11 are used to address implementation issues and challenges faced during development. Finally, the project concludes in week 11 and 12 with the preparation of the timeline and milestones, summarization of findings, presentation of the novelties of work, and a conclusion with supportive remarks, effectively wrapping up the entire process.

CHAPTER 4 - CONCLUSION

4.1 Problem and Motivation

This proposal presented a thorough strategy to implement a cost-effective system for indoor navigation for the VI using QR codes. The project tries to address drawbacks of existing technologies as they are often expensive, and require constant connectivity, despite their potential. Some of the technologies that these companies are using include beacon-based systems, smart glasses and artificial visual recognition. Meanwhile, this approach offers a powerful but cost-effective and easily deployable alternative that is much lighter on computing power.

4.2 Proposed Solution and Implementation

The proposed system integrates the four core modules into a single solution. The QR codes placed at the key locations will store information about that specific spot, such as its location on the prestored building map. When the QR code is scanned, the system uses this information to figure out where the user is. Then, it calculates the best route to the chosen destination and gives step-by-step directions through audio guidance. The system not only guarantees precision and dependability, but its modular architecture also provides flexibility in updating or improving individual parts without affecting the system as a whole.

4.3 Overall Contribution and Impact

To sum up, this proposal presents an implementable, innovative navigational solution supported by solid technical foundations and deep understanding of user needs. It directly tackles the shortcomings as highlighted in previous studies, especially in terms of cost, infrastructure requirements, and ease of deployment. These advantages proves that it is a strong contender for a major advancement in the field of assistive technology for the VI users.

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